Q1) a) what is the value of
Support({apple, pear}) ? explain

b) what is value of
Confidence ({apple, pear} => {orange}) ? explain

c) what is value of
Lift ({apple, pear} => {orange}) ! explain

TID	orange	pear	apple
1	1	1	0
2	0	0	1
3	0	0	0
4	1	1	1
5	0	1	0

Q2) Doc1 = (1,0,2,3) and Doc2 = (2,0,2,4) with is Recent, Rating, Page count, image count

calculate cosine similarity

18

Q3) calculate and construct dissimilarity matrices for

a) nominal data

b) ordinal data

c) symmetric binary data

d) numeric data

E) all data in table

(comment after constructing

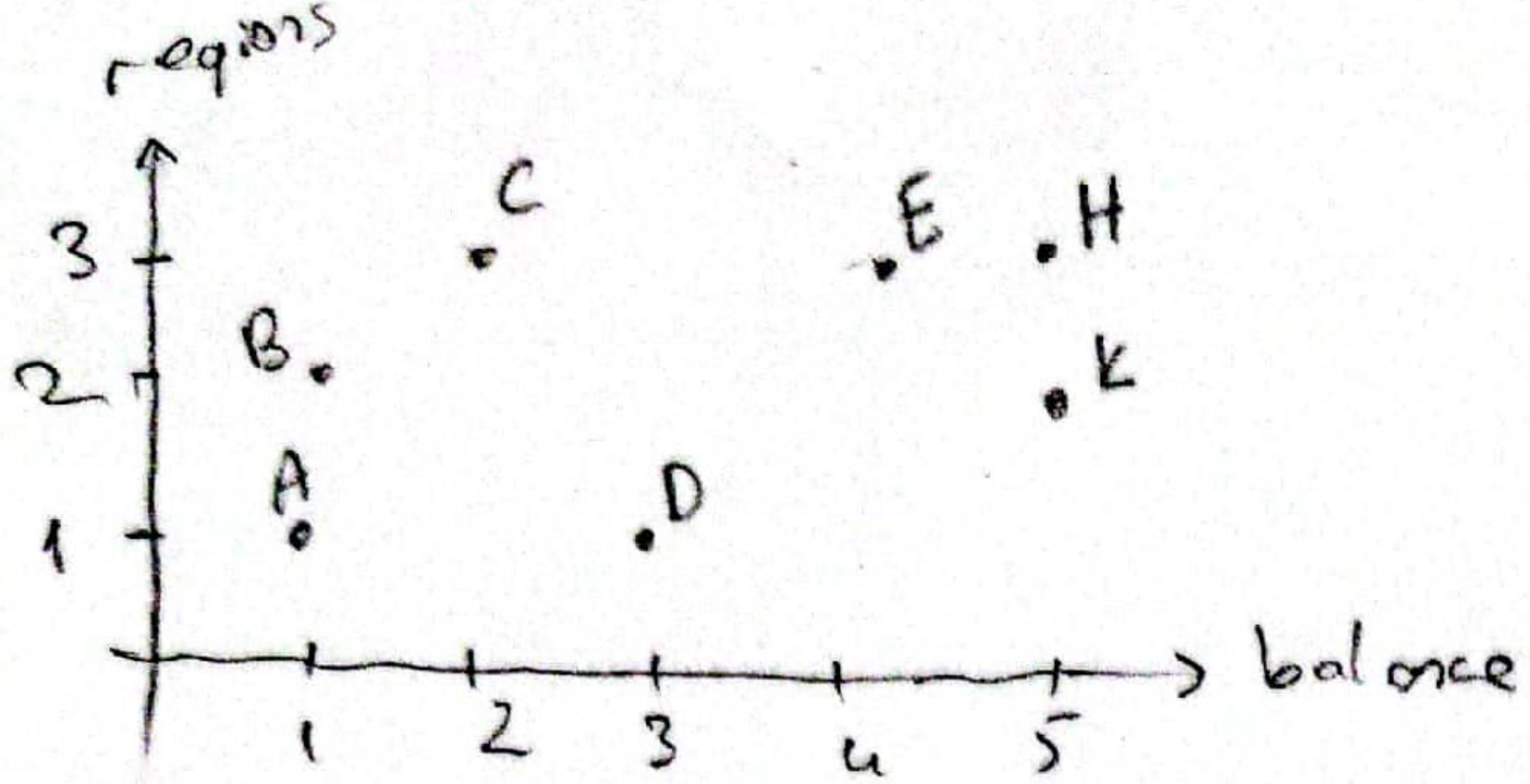
vehicle	color (nominal)	size (ordinal)	needs road (symmetric)	has 4 wheels (symmetric)	lifetime (numeric)
car	black	small	yes	yes	10
train	black	large	yes	No	15
plane	white	medium	No	No	12
SUV	brown	small	yes	ye	6
Van	white	small	yes	yes	5

Use following data for 4,5,6 (balance, regions)

A(1,1), B(1,2) C(2,3), D(3,1), E(4,3), H(5,3), K(5,2)

Q4) use k-medoids with k=2 initially A and B.

(Q5) Agnes with $k=3$, Dendogram ?



(Q6) Use DBScan with minPts=3 and epsilon=1.5

a) show final clusters?

b) which customers are core?

c) " " " border?

d) " " " outliers?